Research on Driving Factors of COVID-19 Protective Behaviours in Australia

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1 Declaration

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2 Abstract

During the COVID-19 pandemic, personal protective behaviour, especially wearing masks, played a crucial role in controlling of the infection. Although public health policies such as mask mandates significantly changed people's compliance, understanding the behavioural driving factors behind protective behaviours are still important. These driving factors are complex and dynamically influenced by macro pandemic factors such as vaccination rates and confirmed cases. This study investigates the determinants of COVID-19 protective behaviours in Australia through a two-phase machine learning framework. In the first phase, this study reproduced Ryan et al.'s (Ryan et al. 2025, 8–15) research, using random forest and XGBoost to classify and predict the protective behaviours before and after the mask mandates. The evaluation results showed that prior to the implementation of the mandates, people's protective behaviour was mainly driven by existing protective habits and frequency of non-household contact. After the mandates, the passage of time (week number) and an individual's willingness to isolate became the main predictive factors. To address the structural limitations of traditional tree models in capturing high-order and nonlinear variable interactions, the upcoming second phase proposes the construction of a Multi-layer Perceptron (MLP) architecture and the integration of state-level weekly vaccination rates and confirmed cases. In addition, to overcome the 'black box' characteristics of neural networks, this study introduced SHAP (SHapley Additive exPlanations) values to quantify feature contributions.

3 Introduction

During the COVID-19 pandemic, individual protective behaviours played a crucial role in controlling the spread of the virus (Guo, Xiang, and Wang 2023, 1–2). Among all protective behaviours, mask wearing was initially recognized as one of the most effective measures [Eikenberry et al. (2020), p.11-13). To cope with future pandemics, understanding the drivers of protective behaviours, particularly mask wearing, is essential for the design of public health policies.

There were many studies investigating why people wear masks from various perspectives (Ryan et al. 2025, 2). Among these, mandated policy is one of the most important factors, which may lead to a sharp increase in the use of masks [Haischer et al. (2020), p.4-9). Recently, Ryan et al. (Ryan et al. 2025, 8–13) investigated the role of face mask mandates in influencing the drivers of protective health behaviours in Australia. However, some key factors have not been considered. For example, the introduction of vaccination may alter individuals' risk perception and behavioural responses (De Nicolò, Villari, and De Vito 2022, 2). There is a need to understand how different external factors drive behaviour change to inform future public health policy-making.

In order to identify these complex relationships, appropriate modelling approaches are needed. Most existing research relies on traditional statistical or tree-based models. Although tree-based models often perform strongly on tabular data, they may be limited in fully capturing complex interactions (Rana et al. 2023, 3). However, research also indicates that for most tabular data, GBDTs(gradient-boosted decision trees) outperform neural networks (NNs). In fact, the comparative performance of these two types of models on tabular data remains a subject of significant academic debate, as results often vary significantly across different datasets. Upon reviewing the literature, there is no existing research applying deep learning to similar datasets within this field.

To address these research gaps, this study integrated vaccination data and COVID-19 case numbers, and will apply deep learning method to find out whether it can capture complex relationships and identify the key drivers of protective behaviours, especially mask wearing.

For the whole study, the primary research questions are:

- How do vaccination rates and COVID-19 case numbers influence the uptake of protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional statistical or tree-based models in predicting protective behaviours?

4 Background

5 Methods

This section details the experimental design and implementation methods of the study.

5.1 Data Sources

To explore the research questions, this study used four datasets. This section introduces the sources and contents of the data. For detailed processing procedures, please refer to the section 5.2.

5.1.1 YouGov Behaviour Tracker Dataset

The Imperial College London YouGov Behaviour Tracker dataset (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020) contains the self-reported survey responses of individuals from 29 countries regarding protective health behaviours against COVID-19 between 2020 and 2022. The survey included demographic and psychological factors (such as gender and age), protective behaviours (such as mask wearing and social distancing), comorbidities, and more features. As this study focuses on the situation in Australia, only the Australian subset of this dataset is used.

5.1.2 Oxford COVID-19 Government Response Tracker Dataset

The Oxford COVID-19 Government Response Tracker dataset (Hale et al. 2021) records the state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023. The recorded policies include containment and closure, economic, healthcare, vaccination, and other general policies. This dataset is used for the mandate period split.

5.1.3 Pandemic Case Data

Reporting of COVID-19 cases varied across each state and territory in Australia. Hence, this study uses a consolidated dataset (covid19data.com.au 2023a) for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. The data records daily confirmed cases at state and territory level granularity, covering the period from January 2020 to August 2023. It is only used in the extension phase.

5.1.4 Vaccination Data

Since the Australian government has not released detailed vaccination data for each state on a weekly basis, this study relies on a vaccine dataset compiled by a third-party team (covid19data.com.au 2023b) to match the timing and locations of the YouGov survey. This dataset collects weekly vaccination totals for each state and territory from official health websites, covering January to August 2022. It is only used in the extension phase.

5.2 Data Cleaning, Variable Construction, and Data Preprocessing

This study preprocessed the datasets based on the methodology of Ryan et al. (Ryan et al. 2025, 3–6) to meet the input requirements of the models. The specific feature conversion and cleaning steps are divided into eight core steps.

5.2.1 Government Mandates

During the COVID-19 pandemic, Australia experienced a variety of enforced orders. Due to the different response measures among local governments, the experience of COVID-19 in each state varied significantly. Therefore, the start time and enforcement of mask mandates varied greatly between different states. In order to evaluate the actual effectiveness of government mandates, this study aligned the data based on the daily mask wearing policy index (0 - `No policy' to 4` -Required at all times') from the OxCGRT dataset (Hale et al. 2021, 6–8).

Following Ryan et al.'s (Ryan et al. 2025, 3) method, and considering the varying levels and effects of policy implementation, this study calculated the 14-day rolling average of each state's index (daily updated 14-day average of each state's index). When the rolling average of a certain state reached 3 for the first time, it was considered that the jurisdiction had actually entered a stable period of mandate intervention. Then, the completion time of the individual survey was compared to their regions, and a variable `during_mandate` was generated for each observation.

5.2.2 Removal of Features with High Missingness

For the YouGov dataset, due to multiple item changes during the tracking period of the survey in Australia, some questions were only asked within an extremely short window of time. To prevent the loss of massive amounts of valid samples during the cleaning phase due to high missingness in a few features, this study counted the number of missing values for each feature, and same as Ryan et al.'s (Ryan et al. 2025, 3) study, set 10,781 as the threshold. Columns with missing values greater than this threshold were considered not in the survey for a long period and were dropped from the dataset.

5.2.3 Repairing Data Gaps Caused by Survey Changes

During the data exploration process, it was found that YouGov modified the informed consent terms of the survey between February 10 and October 18, 2021. This led to a significant structural data gap in health related questions during this period. Deleting these records would mean losing eight months of important data. To this end, this study designed an experiment to fill in missing values, comparing the performance of the model fitted by string filling, retaining `np.nan`, and directly deleting entire rows after processing systematic missing data in this time period, and selecting the appropriate method. All three data sets where tarined and tested on XGBoost, because it is the only model selected in this study that can handle np.nan.

As shown in the figure 1, the experimental results indicate that there is little difference in the performance of the three methods on the training set. Subsequently, this study validated their performance on the test set, and the results were all the same.

As other models in this framework fail to deal with missing values, and deleting these features directly would impair model performance. Finally, this study located this time window using a date mask and filled in these missing values as independent 'N/A' categories, thus preserving valid samples.

5.2.4 Processing of Predictive Variables

The core prediction targets are self protection behaviour and mask wearing. The original survey records the frequency of respondents performing a certain behaviour in the past 7 days. This study

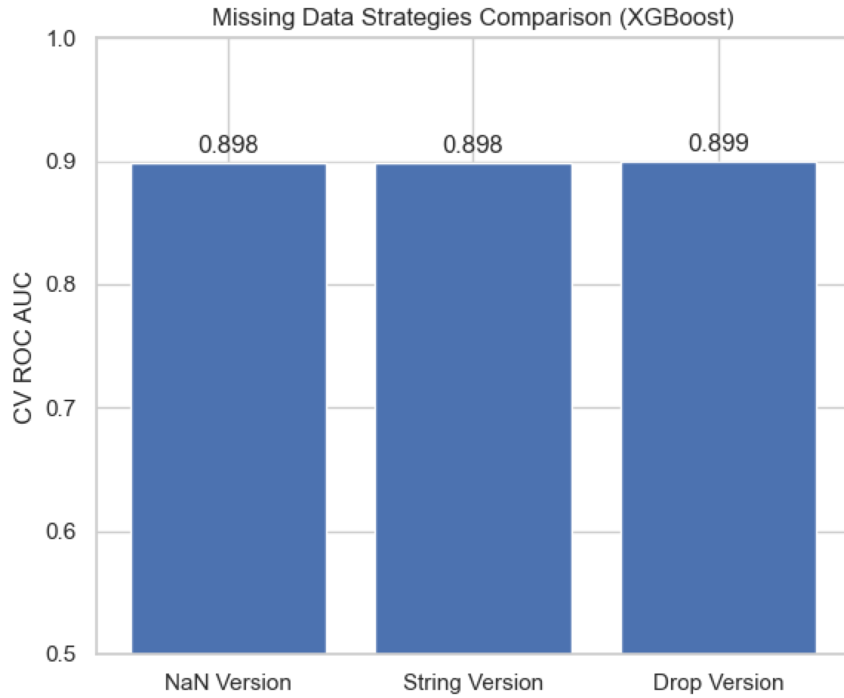

Figure 1: Performance comparison of three imputation methods based on training set cross-validation. Source: generated in $./python_codes/plots.ipynb$

first constructed a dictionary and measured responses on a five-point scale (from 1 - `not at all` to 5 - `always`). Then, features were divided into three logical branches:

- Mask wearing behaviours: There are four specific scenarios related to masks ('Worn a face mask outside your home', 'Worn a face mask inside a grocery store/supermarket', 'Worn a face mask inside a clothing/footwear shop', and 'Worn a face mask on public transportation'). `Mask wearing behaviour` was defined as the median of all scenarios and named `face_mask_scale`. After that, 4 (i.e. `frequently`) was used as the compliance threshold to binarise this feature into a classification label of Yes/No, ensuring a consistent binary outcome definition with the methodology of Ryan et al. (Ryan et al. 2025, 4).
- Protective behaviours: All self protection scenario scores with the prefix `i12_` were extracted, including other protective measures from the survey, such as social distancing, hand washing, and coughing covering. Their overall median was calculated as `protective_behaviour_scale`. The same threshold of 4 or above was used to binarise this faeture into `protective_behaviour_bi`.
- Protective features other than masks: Ryan et al. (Ryan et al. 2025, 10–12) showed that a person's protective behaviours could strongly predict whether they would wear a mask. To avoid confounding effects from respondents' other hygiene habits when predicting mask behaviour, this study specifically excluded the four mask scenarios mentioned above, calculating the median of all remaining protective behaviours as a continuous feature `protective_behaviour_nomask_scale`.

The chart 2 shows the changes in people's protective behaviour from 2020 to 2022 after cleaning

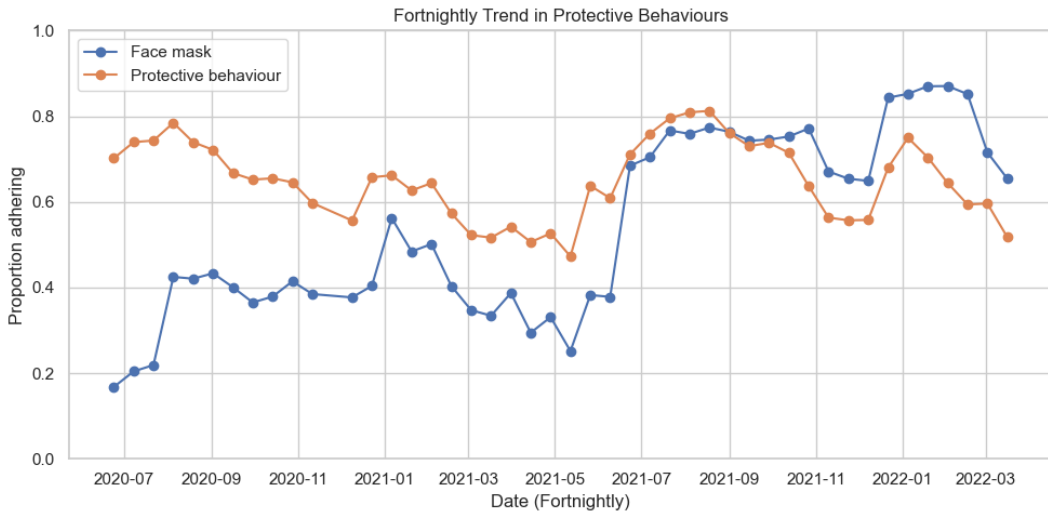

Figure 2: Public compliance between July 2020 and March 2022.

and preprocessing. Initially, the protective behaviours other than masks (orange line) was higher than the proportion of mask wearing (blue line). However, in mid-2021, both classes showed a significant increase. In the later stages, the proportion of people wearing masks exceeded other protective behaviours, reaching a high point of nearly 90%. Both variables exhibit similar temporal trends.

- Other protective predictors: Additionally, this study added three other self-protective features. The first is the average number of non-household contacts `i2_health`, which was retained as a continuous numerical variable. The remaining two are subjective attitude features: willingness to isolate if unwell `i9_health` and willingness to isolate if directed to do so by a healthcare professional `i11_health`. These two subjective attitudes were recorded as categorical responses, and transformed into dummy variables prior to modelling.

5.2.5 Dimensionality Reduction for Sparse Comorbidity Features

In the YOUGOV data, comorbidities were scattered among 13 binary options (from `d1_health_1` to `d1_health_99`), making the data extremely sparse. To improve model efficiency, this study removed all specific disease types and made them into a single variable called `d1_comorbidities`. Respondents who selected 'None of these' were recorded as 'No', 'Prefer not to say' was recorded as 'Prefer_not_to_say', records within the consent gap window were imputed as 'N/A', and any other confirmed diseases were grouped as 'Yes'.

5.2.6 Processing of Demographic Variables

The demographic variables used in this analysis include age, gender (male/female), state (one of the eight states or territories in Australia), employment status, and household size. When dealing with household size, the raw data exhibited a clear long-tail distribution, mixed with text responses. To reduce the effect of extreme values on models, this study assigned a value of 8 to all samples answering '8 people or more'. Invalid samples answering 'I don't know' or 'I don't want to disclose' were directly deleted at the end of the cleaning process.

5.2.7 Processing of Perception, Trust, and Mental Health Variables

In addition to demographic and behavioural data, this study included subjective variables such as perception of illness threat, trust in government, mental health, and wellbeing.

For perception of illness threat (perceived susceptibility and severity), the original responses were measured on a 7-point scale mixed with text (e.g., 7 - 'Agree' and 1 - 'Disagree'). This study cleaned the text characters and converted them into continuous variables. Wellbeing (Cantril Ladder), measured on a 10-point scale, was directly retained as a continuous numerical feature. Finally, for unordered categorical features such as trust in government, PHQ-4 mental health features, and comorbidities, dummy coding was applied with `drop_first = True` to remove the baseline group and prevent multicollinearity.

5.2.8 Vaccination and COVID-19 Case Numbers

This study used the `endtime` of the respondents' survey completion and their `state` value to merge the individual survey records with the corresponding state's vaccination rates and daily new cases.

5.3 Resampling and Validation Pipeline

5.3.1 Class Imbalance and Resampling

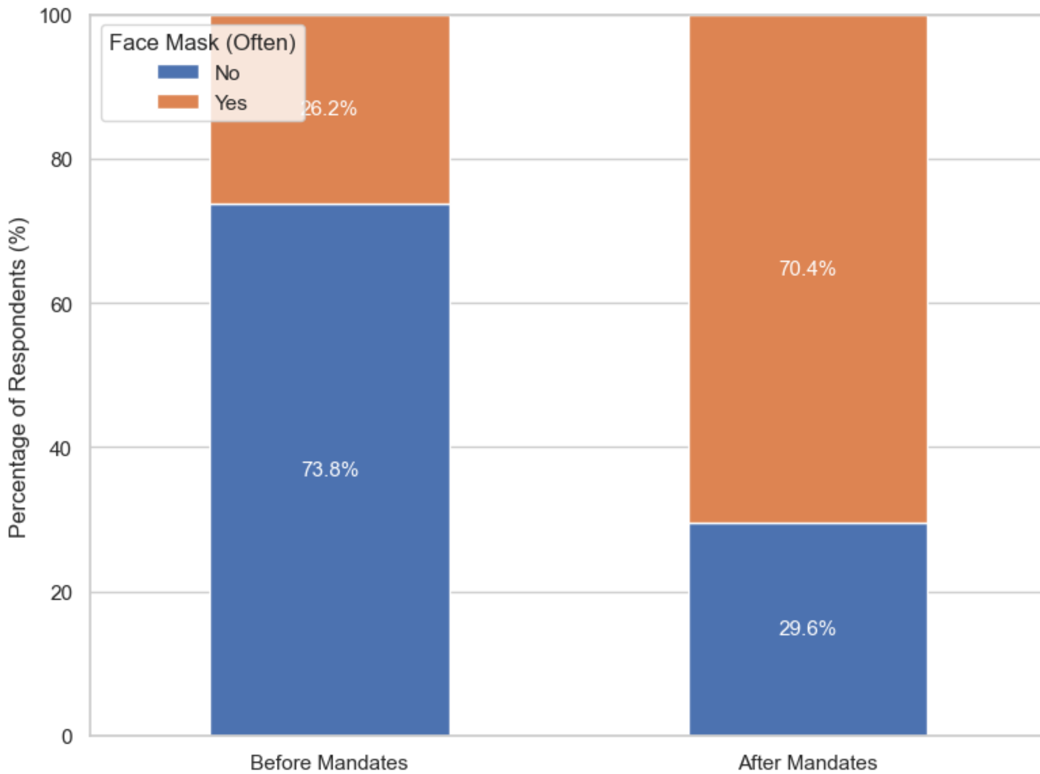

Figure 3: The proportion of respondents frequently wearing face masks before and after the implementation of mask mandates.

As shown in 3, the orange bars represent individuals who frequently wore face masks (Yes), where as the blue bars represent those who did not (No). Before the mandates, the proportion of people who did not wear masks was as high as 73.8%, while those who frequently wore masks accounted for only 26.2%, demonstrating an extreme class imbalance. After mandates, this proportion reversed (70.4% wearing vs. 29.6% not wearing). If such imbalanced data is directly used for training, machine learning models may inherently predict the majority class, resulting in low capture ability for the minority class. To overcome this, this study introduced the random over sampler technique to deal with class imbalance.

5.3.2 Data Splitting and Cross Validation

After completing data cleaning and feature engineering, the data was split into training and testing sets in an 80:20 ratio, stratified by the value of `during_mandate`. To evaluate the effectiveness of policies across multiple dimensions and ensure the reliability of model training, this study constructed a specific data splitting and cross-validation mechanism.

In terms of validation strategy during training, a single validation set is highly susceptible to the randomness of the initial data split. To prevent overfitting during model training, this study implemented a 5-fold cross validation. This mechanism ensures a robust and unbiased evaluation of model performance (Allgaier and Pryss 2024, 1).

Figure 4 shows a simple demonstration of data splitting and cross validation process in this study. Initially, the datase set was divided into a 'training set' and a 'testing set' in an 80:20 ratio. Then, a 5-fold cross validation was performed within the 'training set' (which was split into 5 equal parts) to find the best hyperparameter combination for the model. During each cross validation fold, one part was set aside as a 'Validation Fold' to verify the actual effectiveness of the parameters under this fold. This process was repeated 5 times, and the final result was the average of the five validation results.

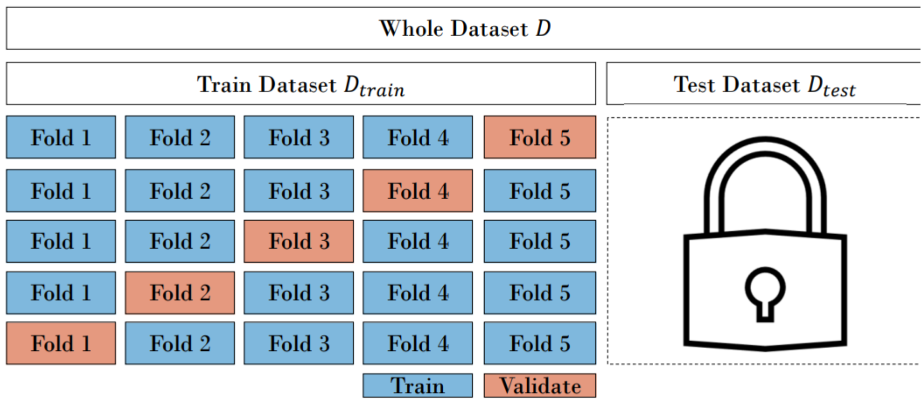

Figure 4: Schematic illustration of a 5-fold cross-validation framework. Source paper: https://www.mdpi.com/2504-4990/6/2/65

5.4 Models

In this chapter, we will introduce all the models specifically used. During the replication phase, this study used logistic regression, classification trees, random forests, and XGBoost. In the expansion phase, MLP was added and random forests was removed.

5.4.1 Logistic Regression

Logistic regression is a generalised linear model widely applied in behavioural analysis, that outputs the probability of a target class. Serving as the baseline model, an L2 regularization penalty was introduced to prevent overfitting in the high-dimensional feature space. Specifically, the inverse of regularization strength (parameter C) was searched across a grid of [0.01, 0.1, 1, 10].

5.4.2 Classification Trees

Classification trees are supervised learning methods that classify data by constructing a series of binary splits based on entropy or Gini impurity (Breiman et al. 1984, 126–30). This model offers high efficiency and interpretability. Joint tuning was performed on the maximum tree depth `max_depth` [5, 10, 15, 20, None], the minimum samples required at a leaf node `min_samples_leaf` [1, 2, 5, 10], and the minimum samples required to split an internal node `min_samples_split`[2, 5, 10].

5.4.3 Random Forests

Random Forest is an ensemble method that employs bagging (bootstrap aggregating) (Breiman 1996, 1–3). Random Forest generates multiple independent decision trees by resampling the training data with replacement while evaluating a random subset of features at each split node. These features ensure that the results are not completely dominated by strong individual classifiers. Hyperparameter tuning focused on the total number of trees `n_estimators` [100, 250, 500], maximum tree depth `max_depth` [10, 20, None], and the maximum number of features `max_features` ['sqrt', 'log2'] to control feature subspace randomness.

5.4.4 XGBoost

XGBoost (Extreme Gradient Boosting) is a highly scalable ensemble boosting algorithm. Unlike the parallel structure in random forests, it fits new trees by leveraging the residual information from previous models to continuously optimise subsequent predictions, a process that defines the boosting mechanism (Chen and Guestrin 2016, 786–89). As a boosting algorithm sensitive to parameter configurations, tuned hyperparameter included `learning_rate` [0.05, 0.1, 0.2], column subsample ratio `colsample_bytree` [0.6, 0.8, 1.0], number of estimators `n_estimators` [100, 250, 500], and maximum tree depth `max_depth` [4, 6, 8].

5.4.5 MLP Model

Due to the fact that the training set contains 35848 observations, it may not be sufficient to train extremely complex neural network models. In the extension work, this study introduced a multilayer perceptron (MLP) model to explore the performance of such deep models in this task. MLP is a class of fully connected feedforward artificial neural network that captures complex non-linear relationships in data.(Hornik, Stinchcombe, and White 1989, 363–65) To optimize the model architecture, this study tuned different network depths by evaluating 1-layer, 2-layer, and 3-layer structures. In each layer, the specific layer size was also tuned. The first hidden layer `module__hidden_size_1` [50, 64, 128] was consistently maintained, while the second layer `module__hidden_size_2` ([0, 16, 25, 50]) and the third layer `module__hidden_size_3` ([0, 8, 16]) included `0` in their search spaces, the value "0" represents that the model did not include this layer at this time, and represents that the model contains less than three or two layers of structure. Additional tuning focused on the learning rate `lr` ([0.0005, 0.001, 0.005]), activation functions `module__activation` ([`nn.ReLU`, `nn.LeakyReLU`, `nn.GELU`, `nn.Tanh`]), the dropout rate `module__dropout_rate` ([0.3, 0.4, 0.5]) to mitigate overfitting, and weight decay `optimizer__weight_decay` ([0.0, 1e-4]) for L2 regularization.

5.5 Model Evaluation and Interpretation

Similar to Ryan et al. (Ryan et al. 2025, 3–8), the reproduction section in Term 1 compared the performance of logistic regression, classification trees, random forests, and XGBoost on the initial dataset. Rather than using `Optuna` in the initial framwork, this study adopted a more intuitive `GridSearchCV` during the hyperparameter tuning stage. This method allowed for the definition of a comprehensive hyperparameter grid prior to tuning. To find the optimal parameter combination, a 5-fold cross validation was applied to the training set, with the grid search explicitly configured to maximise the Area Under the Receiver Operating Characteristic Curve (ROC AUC) as the final refit metric. Since this workflow demonstrated strong performance during reproduction, a similar workflow was created for the extension phase in Term 2, incorporating the MLP model and integrating the cross-validation steps directly into the Imblearn pipeline.

5.5.1 Evaluation Metrics

During the model training and testing phases, multiple performance metrics were recorded.The core evaluation metric of this study is the area under the receiver operating characteristic curve (AUC-ROC).This indicator aims to quantify the comprehensive ability of the model to distinguish between positive and negative categories (i.e. groups that adopt protective behavior and those who do not). An AUC-ROC value approaching 1.0 indicates superior predictive performance.

However, relying only on a single evaluation metric cannot fully reflect the performances of the models. Therefore, this study further introduced Precision, Recall, and F1 Score as auxiliary analysis indicators to provide deeper model insights:

Precision(3)**:** Focuses on evaluating the quality of positive predictions. It represents the proportion of instances that adopted protective behaviors out of all instances predicted as positive by the model.

Recall(4)**:** Focuses on evaluating the model's ability to capture target classification. It measures the proportion of actual positive instances (those who took protective actions) that were successfully classified by the model.

F1 Score(5)**:** In practical applications, there is often an inherent trade-off between precision and recall. The F1 score, as the harmonic mean of the two, provides a balanced single evaluation value to more objectively reflect the overall predictive robustness of the model.

5.5.2 Model Interpretation

In the reproduce phase, feature importance was directly extracted from the best model identified by `GridSearchCV` (where the model was refitted on the entire training set using the best parameters). Since random forest and XGBoost performed best, their built-in `feature_importances`attributes were used.

In the expansion phase, this study aims to obtain more detailed information on the importance of features, especially to clearly see the direction, whether the prediction probability increases or decreases when the features become larger. To achieve this goal, the Shapley Additive exPlans (SHAP) value is introduced.

The core principle of SHAP values originates from the Shapley Value in cooperative game theory (Lundberg and Lee 2017), which views a single prediction of a machine learning model as a 'game' and considers each feature as a 'player'. It achieves fair weight allocation by calculating the "marginal contribution" of each feature to the final prediction result that deviates from the model's average expectation.

Definition: Shapley Value Calculation

The calculation formula for the Shapley value ϕ_i of feature i is defined to ensure a fair distribution of the prediction payout among all features.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \tag{1}$$

Here is the breakdown of the components in Equation (1):

- N: The set of all input features utilized by the model's feature space (e.g., demographic attributes, psychological metrics, and mandate indicators).
- M: The total number of features (i.e., the mathematical cardinality of the feature set N).
- S: A specific coalition (subset) of features formulated prior to the introduction of the target feature i.
- $f_x(S)$: The conditional expectation of the model's prediction for a specific instance x, given only the subset of features S. This represents the model's baseline estimation when the information from the withheld features is integrated out.
- $f_x(S \cup \{i\}) - f_x(S)$: The marginal contribution of feature i to the subset S. This isolates the explicit change in the model's predicted probability attributable exclusively to the inclusion of feature i.
- $\frac{|S|!(M-|S|-1)!}{M!}$: A combinatorial weighting coefficient representing the probability of the feature coalition S forming. It accounts for all possible permutations in which features can be introduced to the model, thereby neutralizing sequence-dependent interaction effects and ensuring a mathematically fair distribution of the prediction payout. [1]

Guarantee local accuracy:

After calculation, SHAP can guarantee local accuracy by expressing the model's prediction for a specific instance as the sum of a base value and the attribution of each individual feature. This relationship is defined as:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{2}$$

The variables in Equation (2) are explicitly defined as follows:

- $f(x)$: The final output (prediction) of the model for a specific data instance x.
- ϕ_0: The base value. This represents the expected model output (the average prediction) over the entire background dataset if no features were known.
- M: The total number of input features for the instance.
- ϕ_i: The SHAP value (attribution) for the i-th feature. This quantifies exactly how much the i-th feature pushes the model's prediction higher or lower than the base value.

This theorem explains exactly how our model arrives at a prediction for any individual respondent. It starts with a baseline average (ϕ_0), which represents the overall expected prediction for the entire dataset. Then, it evaluates the specific impact of each feature for that person (ϕ_i), represented by their individual `shap_values`.

[1] This calculation was generated using the `shap.TreeExplainer()` function on lines 776 - 830 of `ljj_term2.ipynb`, which processes the output from the XGBoost model.

If we take the baseline average and add all the specific SHAP values for a single respondent, the sum will perfectly match the model's final predicted probability $f(x)$. This exact mathematical match is crucial, as it proves that our SHAP analysis reliably explains the model's behavior at the individual level.

6 Results

his section demonstrates and explains the performance of the models, the results of model selection, and the comparison of key predictors.

6.1 Model Performance and Selection

6.2 Comparison of Key Predictors Before and After Mandate Periods

6.2.1 Face Mask Wearing

6.2.2 General Protective Behaviours

7 Discussion

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8.1 Appendix 1: Model Evaluation Metrics

To rigorously evaluate the classification performance of our predictive models, this study ued standard metrics derived from the confusion matrix. The fundamental components of the confusion matrix include True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

Definition: Precision and Recall

Precision quantifies the proportion of positive identifications that were actually correct, whereas **Recall** (also known as Sensitivity or True Positive Rate) measures the proportion of actual positives that were correctly identified by the model.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

Here is the breakdown of the parameters in Equations (3) and (4):

- TP: True Positives (instances correctly predicted as the positive class).
- FP: False Positives (negative instances incorrectly predicted as positive).
- FN: False Negatives (positive instances incorrectly predicted as negative).[2]

Definition: F1 Score and AUC-ROC

The **F1 Score** represents the harmonic mean of Precision and Recall. It provides a singular metric that balances the inherent trade-off between the two, which is particularly useful when optimizing models on imbalanced datasets.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

[2] These evaluation metrics were computed using `sklearn.metrics.precision_score` and `sklearn.metrics.recall_score` on our test set in the evaluation pipeline.